



Cow-level evaluation of plasma essential amino acid clusters and their association with lactating dairy cow performance and efficiency

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ABSTRACT

Studying changes in plasma EAA concentration and profile may provide a foundation for precision protein nutrition strategies for feeding dairy cows. Using data from a systematic review, Letelier et al. (2022a) identified 2 EAA concentration and profile clusters that differed in performance (DMI, daily milk energy output, fat yield and protein yield) and measures of feed and N use efficiency. However, controlled trials are necessary to evaluate these findings with individual cow-level data. We hypothesized that the EAA concentration and profile at the group level obtained from the literature would result in comparable performance and efficiency measures when evaluated at the individual level. Hence, this study aimed to evaluate the clusters identified in the review of Letelier et al. (2022a) using a supervised machine learning model to determine whether the differences in performance and efficiencies between clusters were consistent when assessed with an independent empirical dataset. These data were from a study of 62 cows stratified in 4 lactation stages (early, mid-early, mid-late, and late) fed diets of 13.6%, 15.2%, 16.7%, or 18.3% CP (DM basis) for 8 wk. A random forest algorithm was trained for cluster identification using data from the review published by Letelier et al. (2022a), with Arg, His, Ile, Leu, Lys, Met, Phe, Thr, and Val concentration and profile (i.e., the molar proportion in total plasma EAA concentration) as features and clusters as outputs. A leave-one-trial-out cross-validation approach evaluated the model's classification ability, achieving accuracy, precision, recall, and F₁ score of 84.4%, 87.3%, 85.7%, and 86.5% for cluster 1, and 84.4%, 80.5%, 82.5%, and 81.5% for cluster 2, respectively. The model was then retrained using the entire 62-cow dataset. The EAA concentration and profile were assessed in wk 4 and 8, where 33 and 38 cows were

classified as cluster 1, and 29 and 24 as cluster 2, respectively. Cows in cluster 2 exhibited greater concentrations of Arg (11.8%), His (24.5%), Ile (18.4%), Leu (38.8%), Lys (16%), Met (14.4%), Phe (15.2%), Trp (4.5%), and Val (26.3%) than cows in cluster 1. In addition, cluster 2 cows were determined to have 2.53% greater DMI (28.4 ± 0.23 vs. 27.7 ± 0.22 kg/d) and 2.34% greater ECM yield (42.8 ± 0.51 vs. 43.8 ± 0.54 kg/d), but 2.81% lower N use efficiency ($100 \times$ milk true protein N/N intake; $28.5\% \pm 0.42\%$ vs. $27.7\% \pm 0.44\%$) than cows in cluster 1. There was no difference in feed efficiency and milk component concentration between clusters. These findings confirm that the group-level responses previously observed between EAA-based clusters and measures of dairy cow performance also exist at the individual cow level. However, research with larger datasets is needed to clarify the nature of the relationship between EAA-based clusters and measures of performance, feed efficiency, and N use efficiency.

Key words: plasma EAA concentration, plasma EAA profile, N use efficiency, precision protein nutrition, dairy cow performance

INTRODUCTION

The literature shows that EAA supplementation may increase milk protein production of dairy cows (Haque et al., 2012; Zanton et al., 2014; Yoder et al., 2020). Hence, considering that the federal milk marketing order sets milk price based in part on the quantity of protein delivered (Bozic and Wolf, 2022), protein nutrition and AA supplementation are crucial to the profitability of dairy farms.

Although cows exhibit substantial metabolic flexibility in synthesizing milk protein from various N sources (Johnson et al., 1998; Brito and Broderick, 2007), increasing MP supply by increasing dietary CP without addressing AA requirements can increase urinary N losses and thus raise environmental concerns (Broderick, 2003; Letelier et al. 2022b). This issue has led to growing em-

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

phasis on reducing dietary CP to increase measures of N use efficiency (Erickson et al., 2024a). For precision AA nutrition, Lys and Met are critical in corn-based diets (Lobos et al., 2021; Seleem et al., 2024). Histidine and Leu may also be limited in high-forage or low-protein diets (Varvikko et al., 1999; Schwab and Broderick, 2017). Furthermore, Haque et al. (2012) reported increased milk and protein productions when cows' diets included an ideal combination of Lys, Met, His, and Leu (the 4 most common limiting EAA).

Plasma EAA concentration can provide valuable insights into nutritional strategies aimed at improving milk production and indicators of efficiency. Plasma EAA levels are dynamic and reflect the momentary balance between supply (diet and body protein) and metabolic utilization (Cant et al., 2022). Recently, Daddam et al. (2025) showed that cows with high feed efficiency (measured as residual feed intake) had lower plasma concentrations of most EAA than cows with low feed efficiency. Separately, a meta-analysis of 22 studies including 96 dietary treatments identified 2 plasma EAA clusters that differed in performance (DMI, daily milk energy output, fat yield, and protein yield), indicators of feed efficiency and MP use efficiency (Letelier et al., 2022a). These findings were based on an analysis of both EAA concentration and EAA profile calculated as concentration of EAA (μM) $\times$ 100/concentration total plasma EAA (μM), that is, the molar proportion of an EAA expressed as a percentage. Because the study by Letelier et al. (2022a) relied on treatment means from the literature, it remains unknown whether these group-level results translate to similar responses at the individual cow level.

Supervised machine learning models may offer a promising approach to bridge the gap between group-level findings and individual cow responses of any dietary intervention. By leveraging the structure of decision trees, random forest is a type of ensemble learning method that builds multiple models using randomly selected subsets of the data and predictor variables, aggregating results by taking a majority vote of the predicted class to calculate the final output and improve accuracy (Jiang et al., 2020). It is among the most widely used algorithms for developing predictive models due to its ability to handle complex datasets with numerous variables and interactions (Speiser et al., 2019).

Thus, training a random forest model using the dataset published by Letelier et al. (2022a), with each plasma EAA concentration (μM) and profile (%) as inputs and cluster membership as the output, may enable the classification of individual cows into 1 of 2 clusters. In this study, our hypothesis was that the cluster-based characterization of plasma EAA at the group level determined by Letelier et al. (2022a) would also differ in

performance and indicators of efficiency when evaluated at the individual cow level. Failure to reject this hypothesis would strengthen the contention that the clusters identified in Letelier et al. (2022a) may provide valuable insights into how dietary factors influence plasma EAA concentration and profile of individual cows. To explore this hypothesis, we trained and validated a random forest model to find out whether the differences in group-level performance and efficiencies reported by Letelier et al. (2022a) could be reproduced using individual cow data from the empirical study of Letelier et al. (2022b).

Moreover, to support broader use of the trained model in future research, we aimed to develop a tool to enable classification of cows based on their plasma EAA concentration and profile. This tool could allow researchers to compare performance outcomes between the 2 clusters identified in Letelier et al. (2022a) using the results of future empirical studies with dietary information as input and cluster membership as output. If the cluster classification continues to show consistent relationships with performance and measures of efficiency, the mobile application could provide further insights in nutritional management and precision protein nutrition.

MATERIALS AND METHODS

Cow Management and Dietary Analysis

All procedures involving animals in the study of Letelier et al. (2022b) were approved by the University of Wisconsin–Madison Institutional Animal Care and Use Committee (protocol no. A005043-R01-A03). Briefly, the research conducted at the US Dairy Forage Research Center (Prairie du Sac, WI) included 62 multiparous Holstein cows, stratified by DIM in 4 groups (early, mid-early, mid-late and late lactation). Cows were fed a covariate diet for 2 wk followed by an experimental diet including either 13.6%, 15.2%, 16.7%, or 18.3% CP (DM basis) for 8 wk. Diets were offered once daily as TMR and cows were milked 3 times daily throughout the study. Predictions of MP supply, MP balance, and metabolizable AA supply and utilization efficiency for each CP level and lactation stage are summarized in Supplemental Table S1 (see Notes; NASEM, 2021). Milk production and DMI were monitored daily, and samples for composition were collected from 6 consecutive milkings weekly for analysis with a Foss FT6000 (Foss Electric; Agsource Milk Analysis Laboratory, Menomonie, WI).

Blood samples were collected in wk 4 and wk 8 from the coccygeal vessels of each cow into tubes with K3-EDTA as an anticoagulant. Each week, 6 samples per cow were collected over 3 consecutive days (1345 and 2145 h on d 1, 0645 and 1630 h on d 2, and 0145 and

0945 h on d 3) considering that concentrations of EAA may vary over time (Toledo et al., 2017). At each sampling time, the tubes were kept on ice until all cows were sampled and then centrifuged at $1,500 \times g$ for 15 min at 4°C . The plasma collected from each tube was stored frozen at -20°C . Before analysis, samples were thawed on ice, composited by cow and week in equal volumes, and prepared for AA analysis by liquid chromatography-MS using a stable isotopically labeled standard (Reinhardt et al., 2025). Methionine was considered as the sum of Met and methionine sulfoxide (Mashima et al., 2003), Asn and Asp were summed and reported as Asx, and Glu and Gln were summed and reported as Glx.

Model Training

To evaluate the ability of a supervised learning model to classify the EAA (Arg, His, Ile, Leu, Lys, Met, Phe, Thr, and Val), we conducted a cross-validation analysis using the meta-analysis dataset published by Letelier et al. (2022a). In that study, 2 clusters were identified from a hierarchical analysis on residuals of a mixed-effect model. The present study used these clusters as reference classes to train a supervised learning model as illustrated in Figure 1.

The random forest model was trained using the package “Caret” for R studio version 4.2.1 with the EAA

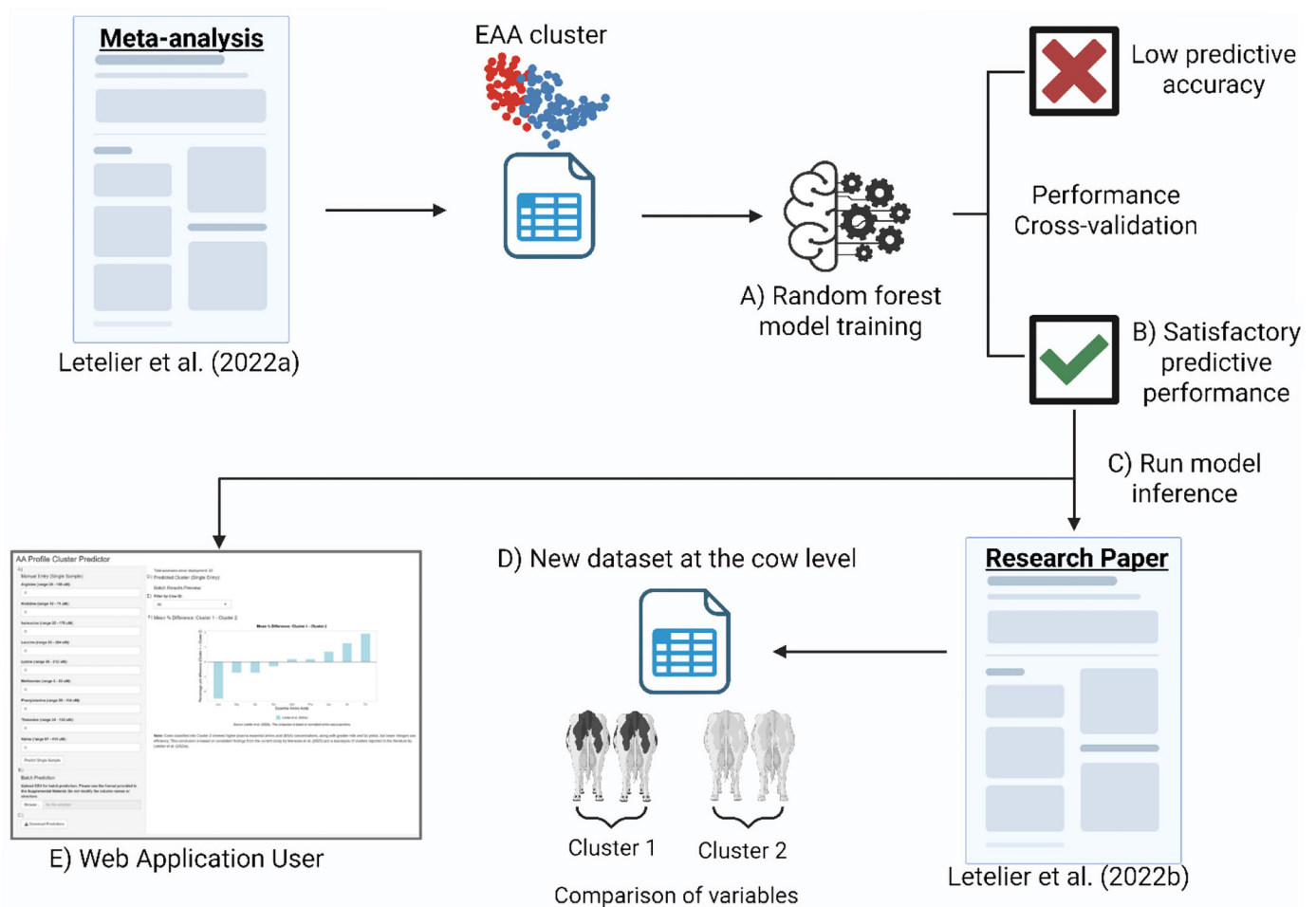


Figure 1. Major steps in EAA cluster modeling and comparison of cow-response variables of this study. (A) Random forest model trained using meta-analysis data to learn the EAA patterns that define each cluster identified in the study of Letelier et al. (2022a). (B) The predictive evaluation of the random forest model was assessed through cross-validation. Once satisfactory results were achieved, the supervised model was considered capable of reproducing the clusters using only plasma EAA information, enabling its use for predicting clusters at the individual level. (C) The trained model predicts cluster membership for individual cows based on their plasma EAA concentration and profile, using data of the study published by Letelier et al. (2022b) to evaluate the comparison of cow-response variables between clusters. (D) Clustered cows are compared for differences in DMI, milk production, milk composition, feed efficiency, and NUE to validate the group-level findings at the individual cow level. (E) A web application tool was developed to enable classification of cows based on their plasma EAA profile in future trials.

concentration and profile, as inputs and the clusters ($n = 2$) as the output. Evaluating cross-validation by treatment may introduce bias due to similarities among treatments within the same study. Therefore, to address potential biases related to study and treatment-within-study effects, 2 cross-validation strategies were applied: leave-one-out cross-validation (**LOOCV**) and leave-one-study-out cross-validation (**LOSOCV**).

We adopted LOOCV because it maximizes the amount of training data in each iteration (i.e., all but 1 treatment), which is particularly advantageous when calibration sets are limited (Bresolin and Dórea, 2020). In contrast, LOSOCV was used to mitigate study-specific bias by withholding all treatments from 1 entire study in each round, which reduces dependence between the training and validation sets (Bresolin and Dórea, 2020). In LOOCV, data from a single treatment (row) were set aside for validation, and the remaining rows were used to train the random forest. In LOSOCV, all data from a single study were excluded for validation, with the remaining studies used for model training. The optimal number of features to consider at each split (mtry) in the random forest was selected using 10-fold cross-validation, with a maximum of 18 features (the total available). To evaluate the predictive quality for classification, the F_1 -score, precision, sensitivity, and accuracy were calculated using the following equations:

$$F_1 = \frac{2TP}{2TP + FP + FN},$$

$$\text{Precision} = \frac{TP}{TP + FP},$$

$$\text{Accuracy} = \frac{TP + TN}{\text{Total}},$$

$$\text{Sensitivity} = \frac{TP}{TP + FN},$$

where TP = true positives, TN = true negatives, FP = false positives, and FN = false negatives.

After completing cross-validation to validate the model's ability to cluster using only concentration and profile information as input, the entire dataset published by Letelier et al. (2022a) was used to train the random forest model. The mtry was selected using 10-fold cross-validation. After training, variable importance was evaluated to identify which features most influenced the model's classification ability. The trained model was used to run inference on a new dataset at the cow level, as described in the next section.

Inference in a New Dataset

The trained model was then applied to infer new clusters for individual cow data stratified in 4 stages of lactation and fed TMR with 4 dietary CP levels as described in Letelier et al. (2022b). Descriptive statistics of EAA for cluster 1 and 2 are presented in Table 1. Descriptive statistics of performance (milk yield, DMI, and ECM) and plasma AA concentrations by cluster for each diet and each stage of lactation are presented in Supplemental Tables S2 and S3 (see Notes). In addition, overall descriptive statistics by cluster for DMI, milk yield, components yield, milk composition, feed efficiency, and N use efficiency (NUE; calculated as $100 \times \text{milk true protein-N/N intake}$) are provided in Supplemental Table S4 (see Notes). Although NEAA were measured, the inferences were based on EAA concentrations and profile because these were the inputs available for the model training. Two rounds of inferences were performed. The first round used blood samples collected at wk 4 (and production and efficiency measures obtained during the first 4 wk of the study), whereas the second round used blood samples collected at wk 8 (and production and efficiency measures obtained from wk 5 to 8 of the study). After identifying which cow belonged to cluster 1 or 2, their individual production and efficiency data were used (a posteriori) to determine the average production and efficiency values of cluster 1 and cluster 2. Repeating the analysis with data collected in wk 4 and wk 8 allowed us to explore the consistency with which the trained model predicted the concentration of individual EAA for each cluster with advancing DIM (see results in Table 1) and the consistency with which cows were assigned to a cluster based on the 2 main factors studied in Letelier et al. (2022b), namely stage of lactation and level of dietary CP.

Furthermore, to account for the large range of EAA profile values among EAA, the percentage difference between clusters in EAA profile was calculated using the following formula: (EAA profile cluster 1 – EAA profile cluster 2)/average EAA profile. Subsequently, mixed models were used to evaluate the effect of clusters on the plasma EAA concentration and profile values, and the corresponding cow production and efficiencies. We tested different models including BW and DIM as covariates and parity as a fixed effect. The model with the lowest Akaike information criterion was used to compare means between clusters.

Statistical Analysis

After cluster prediction, 2 mixed models were tested to compare the differences between clusters 1 and 2. Mixed-effects modeling analyses were performed using

the “lme4” packages for R version 4.2.1. Model [1] was used to compare EAA concentration and profile:

$$Y_{ijk} = \mu + C_i + W_j + (C_i \times W_j) + A_k + e_{ijk}, \quad [1]$$

where Y_{ijk} represents each EAA concentration and profile of interest, μ is the model intercept, C_i is the fixed effect of the i th cluster (1 and 2), W_j is the fixed effect of the j th week (4 and 8), A_k is the random effect of the k th animal ($k = 1$ to 62), and e_{ijk} is the independent identically distributed normal error.

For cow production performance (including DMI, milk yield, milk NEL output, and yields of fat and protein), efficiency (Milk yield/DMI; NEL output/DMI, MP efficiency calculated as milk true protein divided by MP supply), and NUE (calculated as described previously), Model [2] was applied because it includes a covariate period to adjust for individual differences in initial measurements before treatments began, providing a baseline to account for individual variation (Letelier et al. 2022b):

$$Y_{ijklmno} = \mu + Cov_i + C_j + W_k + BW_l + Par_m + DIM_n + (C_j \times W_k) + (BW_l \times W_k) + (Par_m \times W_k) + (DIM_n \times W_k) + A_o + e_{ijklmno}, \quad [2]$$

where $Y_{ijklmno}$ represents the response variable of interest, μ is the model intercept, Cov_i is the effect of the i th covariate ($i = 1$ to 62), C_j is the fixed effect of the j th cluster (1 and 2), W_k is the fixed effect of the k th sampling weeks (1 and 2, corresponding to wk 1 to 4 and wk 5 to 8 of Letelier et al. (2022b), respectively), BW_l

is the fixed effect of the l th BW as a covariate ($l = 1$ to 62), Par_m is the fixed effect of the m th parity, DIM_n is the fixed effect of n th DIM as a covariate ($n = 1$ to 62), A_o is the random effect of the o th animal ($o = 1$ to 62), and $e_{ijklmno}$ is the independent identically distributed normal error.

To be comparable to our results, the performance data reported by Letelier et al. (2022a), were reanalyzed using a model that included study as a random effect, as suggested by St-Pierre (2001; Model [3]). In the original analysis, Letelier et al. (2022a) accounted for study-level variability during clustering by using residuals from a mixed model that included BW, EAA, MP, and study as random effects. However, other performance variables, such as milk yield, DMI, milk composition, and feed efficiency were not modeled with study as random effect. In the reanalysis, no covariate period was included because the variables analyzed represent group averages, and published studies typically do not report covariate values, preventing the use of this information as a covariate:

$$Y_{ij} = \mu + C_i + S_j + e_{ij}, \quad [3]$$

where Y_{ij} represents the performance variable of interest, μ is the model intercept, C_i is the fixed effect of the i th cluster (1 and 2), S_j is the random effect of the j th study ($j = 1$ to 22), and e_{ij} is the independently, identically distributed normal error.

For all models, residual analysis was performed to verify the model assumptions of normality. In interpreting the results $P \leq 0.05$ were considered significant and $0.05 < P \leq 0.10$ were considered a tendency.

Table 1. Descriptive statistics of EAA in cluster 1 and cluster 2 from the trained model applied to the data of Letelier et al. (2022b)

EAA (μM)	Cluster 1						Cluster 2					
	Cows (n)	Mean	Median	SD	Minimum	Maximum	Cows (n)	Mean	Median	SD	Minimum	Maximum
Wk 4												
Arg	33	66.2	64.6	15.4	30.2	95.0	29	76.6	74.8	20.5	26.6	120.5
His	33	26.4	25.5	9.7	12.6	43.0	29	39.1	40.8	10.4	12.4	62.6
Ile	33	83.0	83.2	17.5	34.0	124.4	29	102.5	106.1	19.2	40.2	126.1
Leu	33	119.8	119.5	20.1	55.4	154.5	29	172.9	166.0	25.7	133.9	249.1
Lys	33	77.6	76.9	14.9	36.8	116.6	29	107.9	97.6	34.4	76.0	209.6
Met	33	22.7	20.4	10.4	5.1	45.8	29	28.0	28.5	10.6	5.9	45.8
Phe	33	44.9	45.2	5.7	27.3	64.4	29	53.2	52.9	6.1	42.0	69.0
Thr	33	80.1	81.4	14.8	36.4	108.5	29	87.6	89.6	16.9	62.0	121.9
Val	33	214.7	219.9	45.5	106.2	308.4	29	284.8	282.1	32.9	229.1	362.6
Wk 8												
Arg	38	62.9	65.0	10.9	34.3	83.7	24	69.8	67.4	14.4	47.1	111.6
His	38	28.8	28.7	10.6	11.6	47.4	24	41.1	39.9	9.3	27.6	68.5
Ile	38	85.7	84.7	14.1	64.8	126.3	24	101.3	97.8	13.3	84.2	145.1
Leu	38	120.7	125.2	20.2	62.2	153.6	24	169.8	162.3	26.1	143.0	262.2
Lys	38	77.9	75.3	12.3	59.0	108.0	24	91.6	90.3	19.4	67.6	167.6
Met	38	28.8	27.3	8.7	11.8	51.0	24	30.9	29.8	7.4	21.4	49.6
Phe	38	46.9	44.9	11.7	37.2	111.5	24	53.0	52.5	7.3	44.2	76.1
Thr	38	78.1	76.7	13.6	51.8	112.9	24	79.8	80.7	10.5	62.4	97.5
Val	38	218.4	221.8	44.1	98.5	306.2	24	286.3	279.8	51.4	155.2	412.0

Table 2. F₁ score, recall, precision, and accuracy calculated from Letelier et al. (2022a) data using cross-validation with EAA concentration and profile as inputs

Random forest cross-validation ¹	Cluster	Metric			
		F ₁ score	Recall	Precision	Accuracy
LOOCV	1	90.9	89.3	92.6	89.6
	2	87.8	90.0	85.7	89.6
LOSOCV	1	86.5	85.7	87.3	84.4
	2	81.5	82.5	80.5	84.4

¹LOOCV = leave-one-out cross-validation; LOSOCV = leave-one-study-out cross-validation.

Web Application Development

The AA Profile Cluster Predictor was developed as a web-based application using the Shiny framework (shiny.rstudio.com; Wickham, 2021). Classification of plasma EAA concentration and profile into distinct clusters was performed using the trained random forest model described above. The graphical user interface was constructed using fluidPage components from the Shiny package, with functionality supporting both single-sample predictions through manual entry and batch predictions via CSV file upload (Supplemental File S1, see Notes).

To record the number of users, a local counter is initialized at the start of each session. As explained in the “Model Training” section, the model requires input of the EAA concentrations and profile. Because the EAA profile can be derived from the concentrations, users only need to provide the EAA concentrations, which helps minimize input errors. The input values are validated to ensure that the inference remains within the same range as the training dataset, as machine learning models are limited in their ability to make accurate predictions outside the range of the training data (Meyer and Pebesma, 2021). For batch processing, uploaded CSV files must include the EAA concentrations, from which the profile values (i.e., the molar proportions) are calculated. The results are displayed in the application and made available for download. To aid interpretation, a visualization module was included to compare model-derived differences in EAA profile between clusters against reference values reported by Letelier et al. (2022a).

RESULTS AND DISCUSSION

Cross-Validation

The LOOCV method achieved F₁ score, recall, precision, and accuracy values of 90.9%, 89.3%, 92.6%, and 89.6% for cluster 1, and 87.8%, 90.0%, 85.7%, and 89.6% for cluster 2, respectively (Table 2). Similarly, the LOSOCV method yielded F₁-score, recall, precision,

and accuracy values of 86.5%, 85.7%, 87.3%, and 84.4% for cluster 1, and 81.5%, 82.5%, 80.5%, and 84.4% for cluster 2, respectively. The variable importance resulted from the random forest model are presented in Figure 2, with percentages indicating the relative contribution of each predictor to the model’s classification. Leu (μM) was identified as the most informative variable and was scaled to 100%, with all other variables expressed relative to this reference. For example, Val (μM) showed an importance of ~46.4%, indicating it was approximately half as influential as Leu. In contrast, Met proportion and concentration (μM) exhibited importances of 2.23% and 1.50%, respectively.

Our cross-validation analyses demonstrate that the random forest model successfully reproduced the clustering structure originally generated through hierarchical analysis by Letelier et al. (2022a), regardless of BW or MP effects. Moreover, the consistency of the results when removing 1 study at a time indicates that the clustering was robust and not overly influenced or biased by individual studies. The confusion matrix (Figure 3) illustrates that most of the classes were well balanced, highlighting that the model is not overfitting to any specific class. These results support the application of the trained model for evaluating new cows based on their EAA concentration and profile.

Plasma EAA Concentration and Profile

In our study, each EAA was significantly correlated with all the other EAA (with *P* values ranging from ≤ 0.05 to ≤ 0.001), except for the correlations between Thr and Met and between Thr and Phe (Figure 4). The top 4 strongest correlations we found were between Val and Leu ($r = 0.74$), Val and Ile ($r = 0.73$), Leu and Ile ($r = 0.62$), and Lys and Leu ($r = 0.60$). These 4 correlations were also among the strongest reported in Klein et al. (2013). In addition, the top 2 strongest correlations reported here (i.e., the correlation between Val and Leu and between Val and Ile) were also the top 2 strongest correlations reported in Letelier et al. (2022a). Furthermore, Table 3 presented in Letelier et al. (2022a) agreed with the ab-

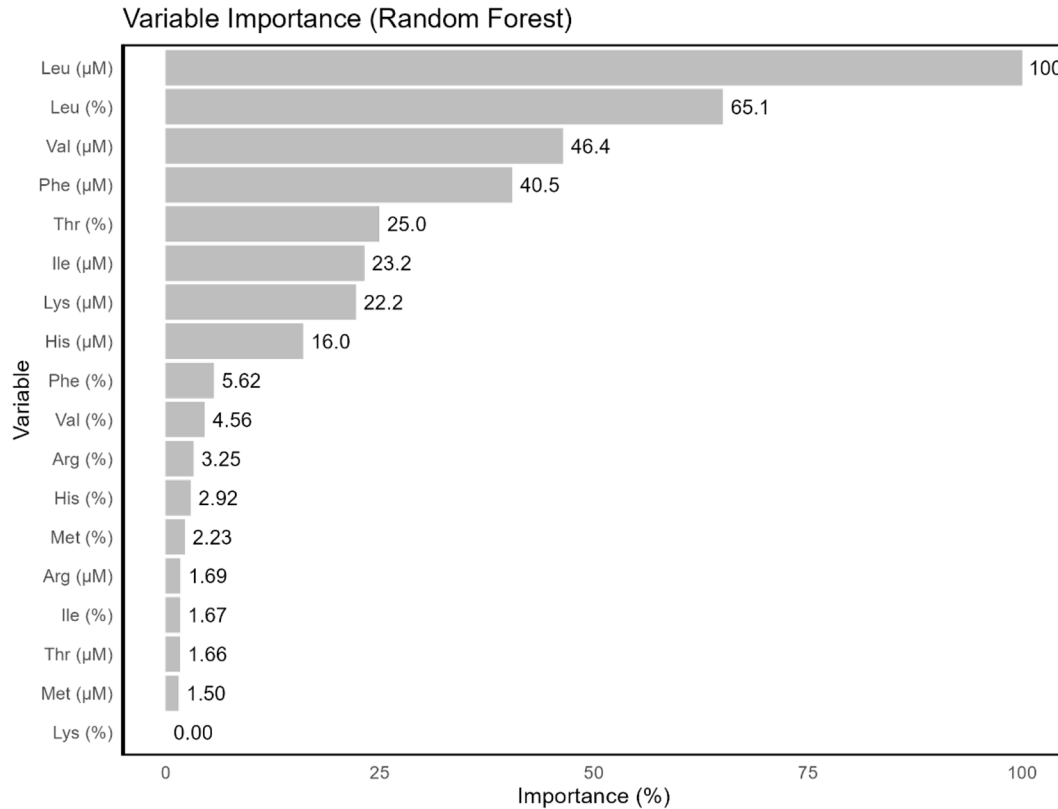


Figure 2. Variable importance for predicting cluster membership based on plasma EAA concentration (μM) and profile (%). The horizontal bars represent the contribution of each variable to the model's ability to classify cows into cluster 1 or 2 based on their plasma EAA concentration and profile. Importance values are scaled such that the most influential variable (Leu [μM]) is set to 100%, with all other variables expressed as a percentage relative to this reference). Longer bars indicate a greater influence on the model's predictive performance.

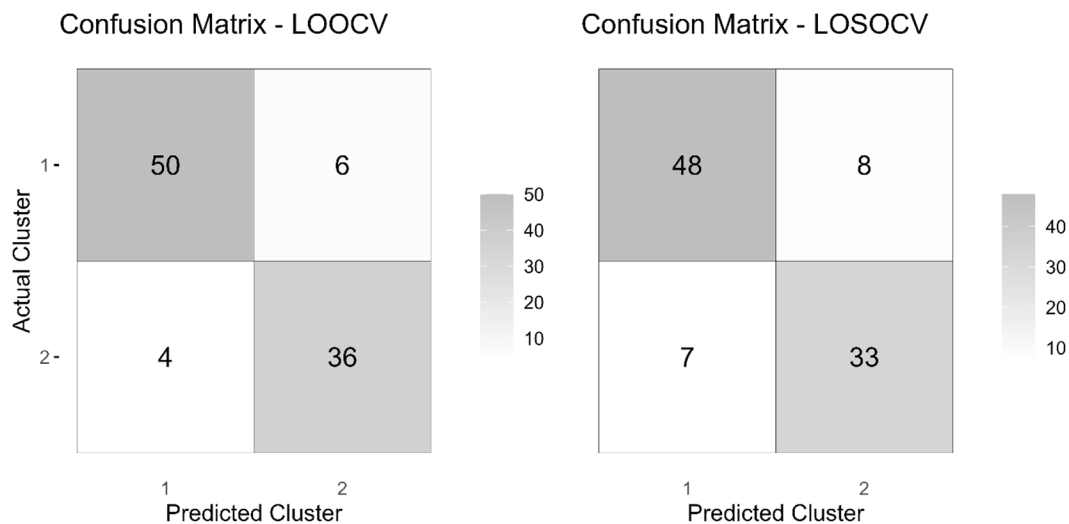


Figure 3. Confusion matrix for binary classification of cluster using leave-one-out cross-validation (LOOCV) and leave-one-study-out cross-validation (LOSOCV). Predicted clusters were determined based on the trained random forest classification algorithm developed in this study, whereas the actual cluster was determined through hierarchical clustering and reported previously in Letelier et al. (2022a). This confusion matrix has a 2×2 structure. The top left cell represents the true positives, where the model predicts cluster 1 and the hierarchical clustering also classifies the cow as cluster 1. The bottom right cell shows the true negatives, where both the model and the reference clustering agree on cluster 2. The top right cell corresponds to false positives, which are cows predicted as cluster 1 by the model but actually classified as cluster 2 in the hierarchical clustering. The bottom left cell shows false negatives, which are cows predicted as cluster 2 by the model but actually belong to cluster 1 according to the hierarchical clustering.

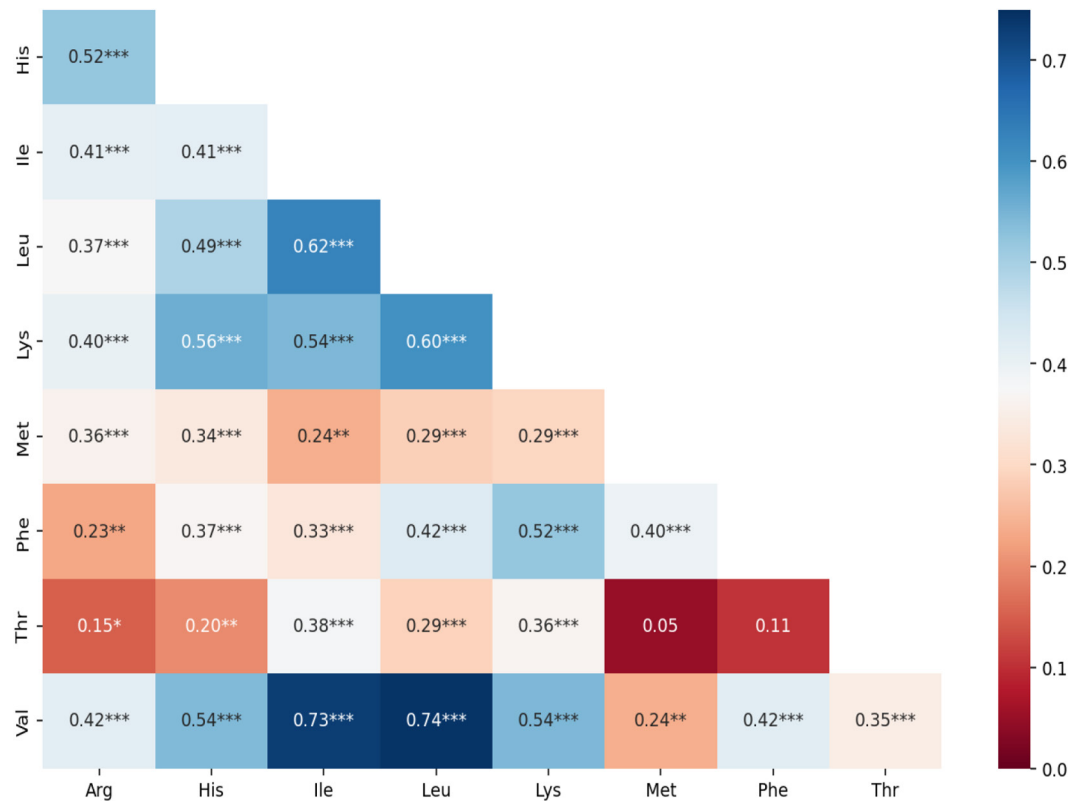


Figure 4. Pearson correlation coefficients among plasma EAA concentration. * $0.01 < P \leq 0.05$; ** $0.001 < P \leq 0.01$; *** $P \leq 0.001$; absence of asterisk(s) indicates $P > 0.05$.

sence of correlation between Thr and Met found here. In contrast to our study, however, Letelier et al. (2022a) found a strong correlation between Thr and Phe and reported no correlation between Met and Ileu.

A brief quantification of clusters by week is described in Table 3. For the plasma samples collected during wk 4 of dietary treatment, as level of dietary CP increased from 13.6% to 18.3%, cows classified into cluster 1 declined and cows classified into cluster 2 increased (Table 3). A similar pattern was observed for the wk-8 plasma EAA samples. The cow distribution between clusters helps explain the greater plasma levels of EAA observed in cluster 2, which is consistent with findings from Patton et al. (2015), who reported an increase in plasma EAA concentrations linearly with duodenal flow. Similarly, Haque et al. (2012) reported higher concentrations of Ile, Leu, Lys, Thr, Val, and total EAA in cows fed diets corresponding to 15.2% CP compared with those receiving 13.6% CP. However, the authors found no differences in Arg concentrations. Arriola Apelo et al. (2014) also reported greater plasma concentrations of Ile and Val in cows fed diets containing 17% CP compared with 15% CP. Likewise, at wk 4, as DIM increased, the number of cows classified into cluster 1 declined whereas the num-

ber of cows classified into cluster 2 increased, although the relationship between cluster and DIM was not present in the samples collected from cows in wk 8. In contrast to the relationship between dietary CP and clusters classification, the biological basis for the temporal change in DIM-associated clustering remains unclear.

Overall, plasma EAA concentrations were lower in cluster 1 cows compared with those in cluster 2 (751 vs. 926 μM ; Table 4). Although Thr showed no significant difference, most AA followed a similar pattern to that reported by Letelier et al. (2022a). In our study, cows in cluster 2 had greater N intake than cows in cluster 1, which may account for these differences. Previous research has shown greater plasma concentrations of Arg, His, Ile, Leu, Lys, Phe, and Val in cows fed diets with greater CP content (Omphalius et al., 2019), corroborating the present findings (Table 4), despite the weak correlations observed between dietary CP level and plasma EAA concentration (Figure 5). However, CP content alone does not fully explain variations in plasma EAA. Brito and Broderick. (2007) evaluated 4 groups of dairy cows fed diets with similar CP concentrations (average 16.6%) but different protein sources, such as urea, solvent soybean meal, cottonseed meal, and canola

Table 3. Distribution of cows by cluster and diet during wk 1 to 4 and wk 5 to 8 of the study of Letelier et al. (2022b)

Treatment ¹	Cows ²	
	Cluster 1	Cluster 2
Wk 4		
13.6% CP diet	13	2
15.2% CP diet	9	7
16.7% CP diet	8	8
18.3% CP diet	3	12
Early lactation	11	4
Mid-early lactation	8	7
Mid-late lactation	8	8
Late lactation	6	10
Total	33	29
Wk 8		
13.6% CP diet	14	1
15.2% CP diet	11	5
16.7% CP diet	6	10
18.3% CP diet	7	8
Early lactation	9	6
Mid-early lactation	10	5
Mid-late lactation	11	5
Late lactation	8	8
Total	38	24

¹Diets 13.6%, 15.2%, and 16.7% were formulated with negative MP balances of 16.3%, 9.70%, and 3.5%, respectively. Diet 18.3% was formulated with a positive MP balance of 5.84%. NEL was positive for all diets. Additional diet details were described by Letelier et al. (2022b). The DIM for each lactation stage was as follows: early, 86 ± 14.9 ; mid-early, 119 ± 10.0 ; mid-late, 167 ± 22.2 ; and late, 239 ± 11.1 .

²Number of cows assigned to cluster 1 or cluster 2 by dietary treatment and lactation stage.

meal, and observed lower plasma EAA concentrations in cows supplemented with urea. These results highlighted the relevance of using not only plasma EAA concentration but also EAA profile to assess the effects of dietary interventions.

The percentage difference in EAA profile between clusters ranged from -1.80% for Leu to $+1.26\%$ for Thr (Figure 6). Except for Arg, the differences in relative molar proportions between clusters 1 and 2 reflected those identified by Letelier et al. (2022a). This outcome was expected, as the EAA profile served as input for training the random forest model. Similar to Letelier et al. (2022a), our aim was not to define an ideal profile, but rather to test whether the EAA clusters derived from group-level data could be applied to individual cows in a controlled setting and still yield comparable insights into performance and N utilization. Similar to Letelier et al. (2022a), our results revealed a significant effect of plasma EAA cluster on selected cow performance variables, measures of feed efficiency, and indicators of N utilization.

In this context, our findings validate the use of plasma EAA clustering using random forest classification at the individual cow level, as the profiles of EAA were very similar to those described by Letelier et al. (2022a).

Cows in cluster 1 and 2 exhibited low but significant correlations ($r = 0.28$) with EAA concentrations and dietary CP. This can be observed in Figure 5, where significant correlations are shown not only for EAA concentration, but also between dietary CP and His and Val in both cluster 1 and cluster 2. The correlation between dietary CP concentration and specific EAA suggests that dietary information could potentially be used to estimate plasma EAA concentration and profile. If these clusters hold true for cow performance and efficiency, future studies should further explore this approach to predict plasma EAA from diet composition, making this framework more applicable in practical settings for farmers and nutrition consultants. Therefore, the following sections compare the performance, feed efficiency, and NUE of cows assigned to cluster 1 compared with those assigned to cluster 2.

DMI and Milk Yield

The dataset published by Letelier et al. (2022a) was reanalyzed using study as a random effect because the study factor was made up of a sample of studies from a larger potential population of studies. When reanalyzed with study as a random effect, many of the productivity associations with cluster were switched from the results reported in Letelier et al. 2022a (Table 5). For example, N intake was 7.79% greater in cluster 2, and milk yield increased for treatment means in cluster 2 compared with cluster 1 (Table 5). In this reanalysis, milk yield increased 2.87% for treatment means classified to cluster 2 ($P = 0.029$), which is consistent with our findings on individual cow data, in which cows in cluster 2 showed a 2.33% increase in milk yield ($P = 0.023$; Table 6).

In the present study, 71.7% of the cows in cluster 2 were fed diets containing 16.7% and 18.3% CP, whereas 66.2% of the cows in cluster 1 were primarily fed diets containing 13.6% and 15.2% CP. Thus, the greater DMI led to higher N intake and milk yield for cows classified in cluster 2, compared with cluster 1 (Table 6), which is in line with results presented by Broderick (2003), who observed linear increases in these variables when dietary CP increased from 15.1% to 18.4%. Furthermore, the study by Omphalius et al. (2019) indicated that increasing dietary CP from 12.9% to 19.9% influenced DMI depending on the level of energy in the diet, but increased milk yield regardless of the level of energy and increased plasma concentration for Arg, Ile, Leu, and Phe.

Dietary AA supplementation may influence the performance of dairy cows (Schwab and Broderick, 2017). Increasing the CP content of the diet often is related positively to DMI (Allen, 2000). In our study, cows with an EAA concentration and profile classified as cluster 2 exhibited a 2.53% increase in DMI ($P = 0.001$) and a 2.57%

Table 4. Comparative analysis of observed plasma EAA and NEAA concentrations between cluster 1 and cluster 2; values presented as mean $\pm$ SE

AA ¹ (μ M)	Cluster 1 (n = 71)	Cluster 2 (n = 53)	P-value		
			Cluster	Week	Cluster $\times$ week
Arg	65.4 $\pm$ 2.0	73.1 $\pm$ 2.3	0.006	0.004	0.305
His	29.8 $\pm$ 1.4	37.1 $\pm$ 1.6	<0.001	0.086	0.462
Ile	85.3 $\pm$ 2.1	101 $\pm$ 2.4	<0.001	0.742	0.481
Leu	121 $\pm$ 2.9	168 $\pm$ 3.3	<0.001	0.940	0.943
Lys	79.4 $\pm$ 1.8	92.1 $\pm$ 2.1	<0.001	0.038	0.092
Met	25.7 $\pm$ 1.1	29.4 $\pm$ 1.3	0.034	0.008	0.352
Phe	45.4 $\pm$ 0.6	52.3 $\pm$ 0.8	<0.001	0.293	0.449
Thr	80.0 $\pm$ 1.9	83.6 $\pm$ 2.1	0.177	0.001	0.230
Val	224 $\pm$ 5.4	283 $\pm$ 6.1	<0.001	0.470	0.759
Gln	108 $\pm$ 4.5	114 $\pm$ 5.1	0.369	0.111	0.758
Gly	299 $\pm$ 9.1	259 $\pm$ 10.4	0.003	0.065	0.810
Ala	259 $\pm$ 4.8	274 $\pm$ 5.3	0.020	0.055	0.262
Pro	91.0 $\pm$ 2.1	82.6 $\pm$ 1.9	0.002	0.928	0.978
Tyr	54.9 $\pm$ 1.1	61.5 $\pm$ 1.2	<0.001	0.855	0.450
Cys	57.2 $\pm$ 2.2	60.8 $\pm$ 2.6	0.303	0.039	0.985
Trp ²	52.9 $\pm$ 0.85	56.5 $\pm$ 0.96	0.003	0.003	0.866
TMG	30.8 $\pm$ 1.8	35.4 $\pm$ 2.0	0.044	<0.001	0.068
3-MH	5.43 $\pm$ 0.26	4.73 $\pm$ 0.29	0.063	0.577	0.895
Asp	19.1 $\pm$ 0.91	21.6 $\pm$ 1.06	0.089	0.391	0.139
Glu	168 $\pm$ 3.32	165 $\pm$ 3.78	0.526	<0.001	0.751
Asn	15.2 $\pm$ 0.70	16.1 $\pm$ 0.80	0.370	0.828	0.156
Ser	114 $\pm$ 2.95	114 $\pm$ 3.38	0.977	0.094	0.428
Asx	34.8 $\pm$ 1.25	37.9 $\pm$ 1.45	0.102	0.850	0.972
Glx	279 $\pm$ 4.79	282 $\pm$ 5.42	0.697	0.231	0.541
G1 ³	99.1 $\pm$ 2.5	121 $\pm$ 2.8	<0.001	0.027	0.486
G2 ⁴	650 $\pm$ 13.5	805 $\pm$ 15.5	<0.001	0.345	0.309
BCAA ⁵	427 $\pm$ 10.0	551 $\pm$ 11.4	<0.001	0.863	0.743
EAA ⁶	751 $\pm$ 15.0	926 $\pm$ 17.2	<0.001	0.696	0.290
NEAA ⁷	1,591 $\pm$ 22.3	1,611 $\pm$ 25.4	0.530	0.021	0.769

¹TMG = Trimethylglycine; 3-MH = 3-methylhistidine; Asx = sum of Asn and Asp; Glx = sum of Glu and Gln.

²Trp is an EAA but it was not included in this analysis to maintain consistency with the analyses published by Letelier et al. (2022a).

³G1 = group 1 of EAA; includes His, Met, and Phe (Trp was not included).

⁴G2 = group 2 of EAA; includes Arg, Lys, Ile, Leu, Thr, and Val.

⁵BCAA = branched-chain AA; includes Ile, Leu, and Val.

⁶EAA = Arg, His, Ile, Leu, Lys, Met, Phe, Thr, and Val (Trp was not included).

⁷NEAA = Gln, Gly, Ala, Plo, Tyr, Cys, TMG, 3-MH, Asp, Glu, Asn, Ser, Asx, and Glx.

increase in N intake ($P = 0.019$) compared with cows in cluster 1 (Table 6). Similarly, the dataset reported by Letelier et al. (2022a) reanalyzed using study as random effect showed greater N intake for treatment means from cluster 2, although no difference in DMI was detected between clusters in the reanalysis (Table 5). The increase in AA concentration in diets (Supplemental Table S1) may enhance the clearance of metabolic fuels such as acetate and propionate, thereby stimulating DMI (Allen, 2000). In addition to DMI, the type of metabolic fuel is another factor that may influence the plasma concentration of EAA (Danes et al., 2020).

The role of individual EAA has also been emphasized in literature. Methionine and Lys, often considered the most limiting AA in dairy cows, have been shown to improve milk production, FCM, and milk protein when supplemented alone or in combination (Wang et al., 2010). In the present study, Met and Lys were greater

in cluster 2, however, in the proportion of the total EAA only Lys was significantly greater in cluster 2. Cows in cluster 2 also had greater milk yield, ECM, and fat and lactose yields but similar milk composition compared with cows in cluster 1 (Table 6). These findings reinforce the idea proposed by Letelier et al. (2022a) that the relative proportion of the EAA in circulation (i.e., the EAA profile) is critical to cow performance. In contrast, despite greater milk yield of cows in cluster 2 than in cluster 1, no differences in feed efficiency variables were observed between clusters in our study.

Milk Component Yield and Composition

The observed improvements in DMI and milk yield in cluster 2 suggest that plasma EAA concentration and profile can influence metabolic processes relevant to milk synthesis. Because the mammary gland is a major

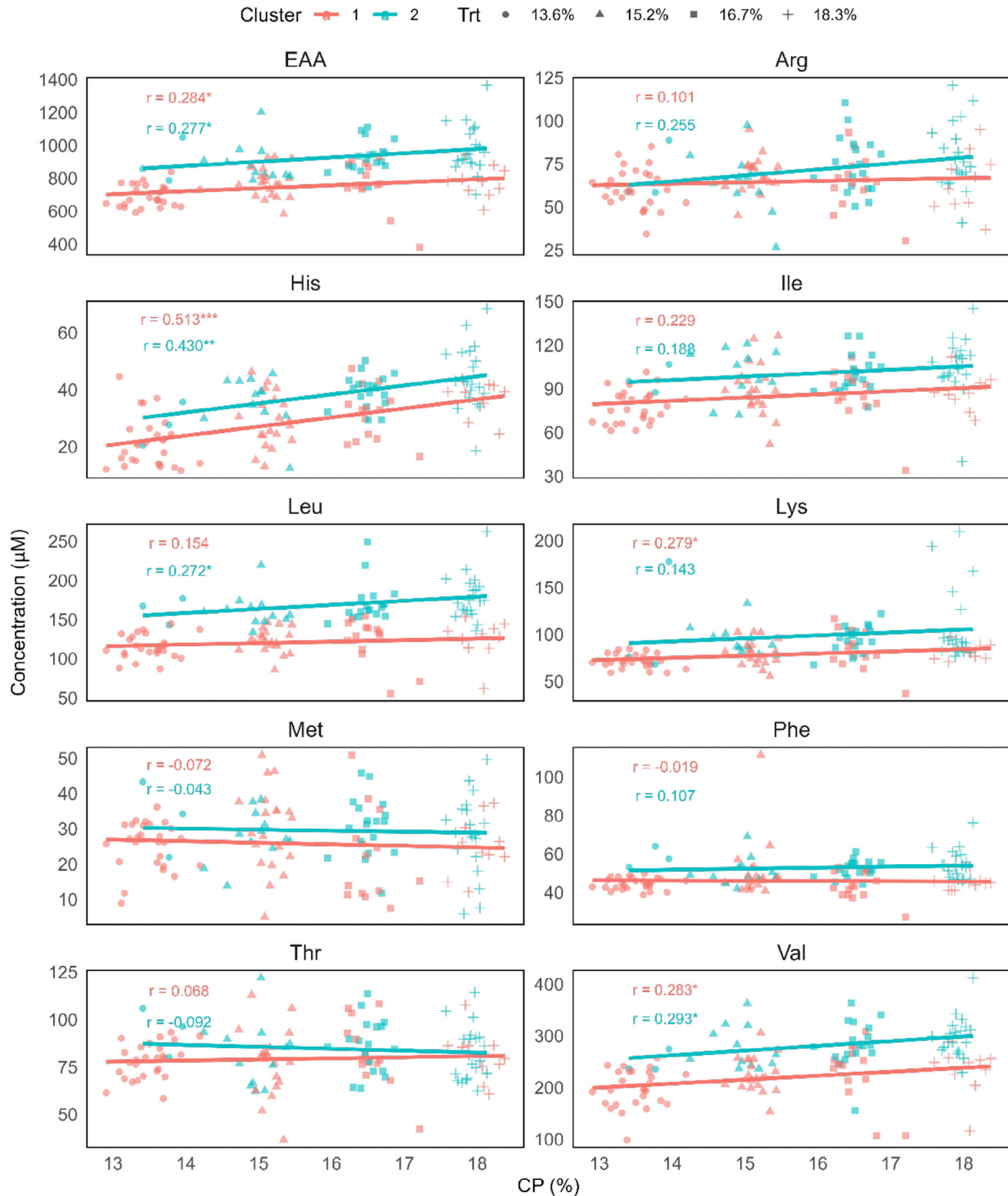


Figure 5. Correlations between dietary CP and plasma EAA. The CP was calculated as $(N \text{ intake} \times 6.25)/\text{DMI}$. This figure displays the Pearson correlation coefficients (r) between dietary CP content and plasma concentrations of total and individual EAA across 2 clusters. Each subplot represents a specific AA, with colored points and lines corresponding to cows from 2 different clusters. Red and blue lines represent linear trends for cluster 1 and cluster 2, respectively. Symbols indicate dietary CP levels: circles = 13.6%, triangles = 15.2%, squares = 16.7%, and plus signs = 18.3%. The 13.6%, 15.2%, and 16.7% CP diets were formulated with negative MP balances of 16.3%, 9.70%, and 3.5%, respectively. The 18.3% CP diet was formulated with a positive MP balance of 5.84%. The NEL was positive for all diets. * $P \leq 0.05$; ** $P \leq 0.01$; *** $P \leq 0.001$.

Table 5. Comparative analysis of performance by cluster with study as random variable (reanalyzed results from Letelier et al., 2022a; means $\pm$ SE)

Item	Cluster 1 (n = 56)	Cluster 2 (n = 40)	P-value
Cow performance			
DMI, kg/d	23 $\pm$ 0.78	23.3 $\pm$ 0.79	0.169
N intake, g/d	578 $\pm$ 19.3	623 $\pm$ 20.9	0.010
Milk yield, kg/d	34.8 $\pm$ 1.48	35.8 $\pm$ 1.50	0.029
Milk NEL, ¹ Mcal/d	24.2 $\pm$ 0.89	24.7 $\pm$ 0.91	0.189
Milk components yield, g/d			
Fat	1,260 $\pm$ 48.4	1,304 $\pm$ 49.5	0.043
Protein	1,069 $\pm$ 40.2	1,095 $\pm$ 40.7	0.065
Lactose	1,642 $\pm$ 73.3	1,687 $\pm$ 74.1	0.042
Milk composition, %			
Protein	3.71 $\pm$ 0.12	3.74 $\pm$ 0.13	0.592
Fat	3.16 $\pm$ 0.05	3.16 $\pm$ 0.05	0.888
Lactose	4.8 $\pm$ 0.04	4.78 $\pm$ 0.04	0.288
Feed efficiency			
MY:DMI ²	1.52 $\pm$ 0.03	1.54 $\pm$ 0.03	0.245
NEL:DMI ³	1.06 $\pm$ 0.03	1.07 $\pm$ 0.03	0.763
MP efficiency ⁴	46.5 $\pm$ 0.834	44.2 $\pm$ 0.987	0.090
NUE ⁵	29.8 $\pm$ 0.968	28.0 $\pm$ 1.014	0.004
PUN, ⁶ mg/dL	11.8 $\pm$ 0.838	14.3 $\pm$ 0.905	<0.001

¹NRC (2001).²MY = milk yield; MY (kg/d):DMI (kg/d).³NEL (Mcal/d): DMI (kg/d).⁴Milk true protein/MP supply.⁵N use efficiency = true protein-N:N intake.⁶Plasma urea N.

site of EAA utilization for milk component production, it is important to evaluate whether differences in EAA concentration and profile are also reflected in milk composition. Milk protein yield is known to increase in response to greater supplies of NEL or MP, as shown by

Broderick (2003), Lapierre et al. (2012), and NASEM (2021). Supplementing for a balanced EAA concentration and profile has also been linked to improvements in milk protein yield, as demonstrated in studies by Haque et al. (2012) and Yoder et al. (2020). The uptake of group 1 EAA (i.e., His, Met, and Phe) by the mammary gland generally mirrors their composition in milk protein (Lapierre et al., 2012), whereas group 2 EAA are typically taken up in excess of their secretion in milk protein. In our study, both group 1 and group 2 EAA were present at greater concentrations in cluster 2 compared with cluster 1 (Table 4). However, the overall profiles of groups 1 and 2 were similar between the clusters (Figure 6).

Although plasma EAA concentration and profile reflect the supply and demand and do not directly indicate mammary gland uptake, cows classified in cluster 2 showed a trend toward greater milk protein yield ($P = 0.116$). A similar tendency ($P = 0.065$) was observed when the database of Letelier et al. (2022a) was reanalyzed to include study as a random effect (Table 5). Although greater plasma EAA concentrations often lead to increased milk protein synthesis, the levels of some group 1 AA, such as Met and Phe (Lapierre et al., 2012), were similar across clusters (Figure 6). These concentrations were also similar to those reported by Haque et al. (2012) in cows with balanced AA profiles and comparable milk production. Therefore, future studies could further evaluate EAA cluster patterns to determine whether there is a relationship between clusters with mammary gland uptake, or whether the observed trend toward greater milk protein yield reflects the greater MP allowance in diets containing 16.7% and 18.3% CP (where most cluster 2 cows were

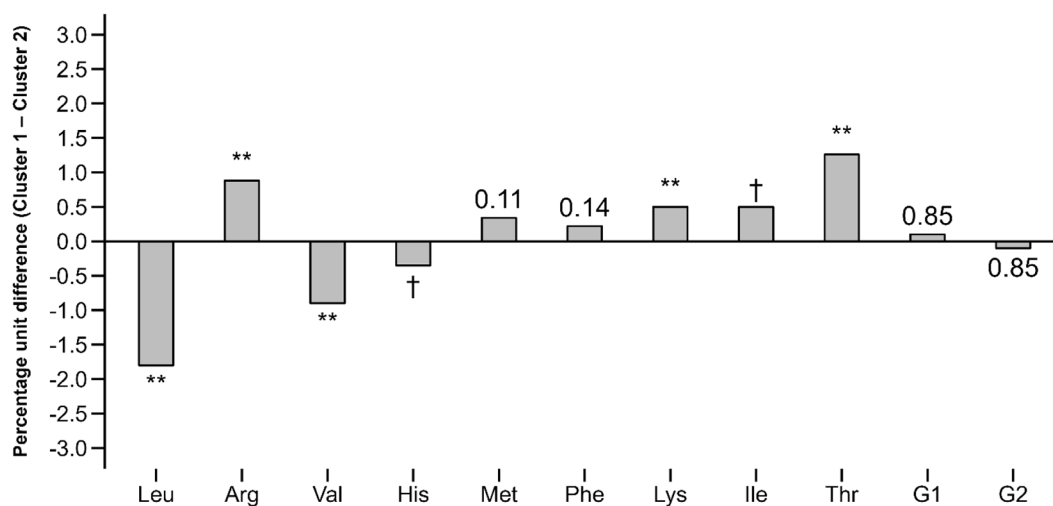
**Figure 6.** Difference in the relative molar proportion of individual plasma EAA concentration as described by Letelier et al. (2022a; % of total plasma EAA concentration) for cows in cluster 1 and cluster 2. G1 = group 1 of EAA; includes His, Met and Phe (Trp was not included for consistency with Letelier et al., 2022a) G2 = group 2 of EAA; includes Arg, Lys, Ile, Leu, Thr, and Val. † $P \leq 0.10$; ** $P \leq 0.01$; P -value shown above or below bar when $P > 0.1$.

Table 6. Cow characteristics, production performance, and efficiencies of cows in cluster 1 and cluster 2; values presented as mean $\pm$ SE

Variable	Cluster 1 (n = 71)	Cluster 2 (n = 53)	P-value ¹						
	C	Cov	W	BW	DIM	Par	W $\times$ C	W $\times$ BW	W $\times$ DIM
Cow performance									
DMI, kg/d	27.7 $\pm$ 0.22	28.4 $\pm$ 0.23	<0.001	<0.001	<0.001	0.907	0.843	0.036	0.253
N intake, g/d	700 $\pm$ 12.9	718 $\pm$ 13.10	<0.001	0.050	0.139	0.400	0.392	0.015	0.959
Milk yield, kg/d	38.6 $\pm$ 0.46	39.5 $\pm$ 0.48	<0.001	<0.001	0.130	0.014	0.260	<0.001	0.648
ECM, kg/d	42.8 $\pm$ 0.51	43.8 $\pm$ 0.54	<0.001	<0.001	0.035	0.017	0.159	<0.001	0.546
Milk component yield, g/d									
Fat	1,585 $\pm$ 21	1,639 $\pm$ 22	<0.001	<0.001	0.013	0.026	0.060	0.002	0.333
True protein	1,243 $\pm$ 16	1,265 $\pm$ 17	<0.001	<0.001	0.009	0.033	0.123	<0.001	<0.001
Lactose	1,820 $\pm$ 23	1,874 $\pm$ 24	<0.001	<0.001	0.057	0.010	0.309	<0.001	0.896
Milk composition, %									
Fat	4.18 $\pm$ 0.03	4.17 $\pm$ 0.03	<0.001	<0.001	0.326	0.185	0.721	0.012	0.632
True protein	3.26 $\pm$ 0.02	3.24 $\pm$ 0.02	<0.001	<0.001	0.006	0.175	0.008	<0.001	<0.001
Lactose	4.71 $\pm$ 0.01	4.72 $\pm$ 0.01	<0.001	0.676	<0.001	0.096	0.343	0.001	0.067
MUN, mg/dl	11.4 $\pm$ 0.37	11.4 $\pm$ 0.38	0.232	0.003	0.659	0.198	<0.001	0.494	0.117
F:P ratio ²	1.28 $\pm$ 0.01	1.29 $\pm$ 0.02	0.318	<0.001	0.030	0.160	0.150	0.219	0.013
Feed efficiency									
MY:DMI ³	1.39 $\pm$ 0.02	1.39 $\pm$ 0.02	<0.001	<0.001	0.104	<0.001	0.173	0.023	0.799
NEL:DMI ⁴	1.04 $\pm$ 0.01	1.04 $\pm$ 0.01	<0.001	<0.001	0.009	<0.001	0.126	0.025	0.161
NUE ⁵	28.5 $\pm$ 0.42	27.7 $\pm$ 0.44	<0.001	0.094	0.387	0.004	0.054	0.025	0.005

¹Cov = Covariate period; W = weeks (1–4 and 5–8); Par = parity; C = cluster.²True protein content (%):fat content (%).³MY = milk yield; MY (kg/d):DMI (kg/d).⁴Milk NEL (Mcal/d):DMI (kg/d).⁵N use efficiency = true protein-N:N intake.

concentrated, as 72% of cows in cluster 2 were fed 1 of these 2 diets) compared with diets containing 13.6% and 15.2% CP (where most cluster 1 cows were concentrated, as 66% of cows in cluster 1 were fed 1 of these 2 diets).

Cows in cluster 2 produced 3.41% more milk fat compared with those in cluster 1. The literature shows mixed findings on the relationship between milk fat yield and dietary protein level or AA supplementation. Some studies reported no significant changes in milk fat yield following AA supplementation (Appuhamy et al., 2011; Yoder et al., 2020), whereas others documented significant effects or tendencies (Broderick, 2003; Wang et al., 2010). Previous studies have reported that Met supplementation can increase milk fat production (Zanton et al., 2014), although component production effects may be closely related to the observed increases in both DMI and milk production responses between clusters because milk composition did not differ between the clusters.

Nitrogen Use Efficiency

In this study, NUE was greater for cows classified in cluster 1 compared with those classified in cluster 2 (28.5% vs. 27.7%, $P = 0.033$; Table 6). Protein supplied in excess of requirement is metabolized at an energy cost (Reed et al., 2017) and leads to increase N excretion (Brito and Broderick, 2007). In the present study, cows classified in cluster 2 had, on average, a numerically small but significantly greater ($P = 0.019$) N intake (due mainly to greater DMI) than cows classified in cluster 1 (Table 6). This differential N intake may have contributed to the difference in NUE between the 2 clusters as NUE is typically reduced when dietary N intake increases (Castillo et al., 2000; Erickson et al., 2024b). Overfeeding N can affect milk and consequently milk true protein yield (Ipharraguerre and Clark, 2005). However, we speculated that in this study, the supply of dietary EAA associated with distinct protein supplements used to formulate the dietary treatments was more important than N intake in explaining the difference in milk production between clusters (Table 6). Thus, future studies should evaluate further the possible causes of greater NUE observed in this study. The tool described in the following section could support such studies by predicting cluster membership in new trials and facilitating the evaluation of EAA cluster effects on performance and indicators of NUE in dairy cows.

Web Application User

Our study is the first to use empirical cow performance data to evaluate the EAA clusters proposed by Letelier et al. (2022a). These authors highlighted the need for further research to test the robustness of using plasma

EAA profile as a holistic indicator of the EAA composition in MP. However, applying these clusters in new studies requires retraining a supervised machine learning model. To address this limitation, we developed a tool based on the model trained and validated in the present study. This tool enables researchers to classify individual cows according to literature-derived EAA clusters without the need to retrain a model, facilitating standardized comparisons across studies. It is important to emphasize that we are not proposing the EAA profile identified by Letelier et al. (2022a) as an ideal reference. Rather, we demonstrate that group-level EAA clusters can effectively classify individual cows and reveal consistent differences in performance, supporting broader validation and application of EAA clustering in future research.

All functionalities of the tool's user interface are illustrated in Figure 7. The model can be accessed at <https://precisiondigital2025.shinyapps.io/AAClusterPredictor/>. Users can obtain a prediction for a single cow just as an illustrative example by manually entering plasma concentrations of 9 EAA to classify the cow into one of the clusters. To evaluate an entire trial against the clusters identified by Letelier et al. (2022a), users can use the batch prediction feature by uploading a CSV file containing EAA concentration for multiple cows (see Supplemental Material S1, see Notes). After prediction, users can either download the batch results as a CSV file or visualize the average percentage differences in normalized EAA profile between clusters directly within the application.

It is important to note that this tool was developed using data from 22 studies and 96 treatment means. As a result, any input values falling outside the ranges (Meyer and Pebesma, 2021) shown in Figure 7 should be considered out of range, and predictions for such values may be unreliable or lead to incorrect cluster classification based on the greatest similarity to cluster 1 or 2. Additionally, the tool requires EAA plasma concentrations as input. Including dietary AA concentrations will not provide accurate predictions, as differences in rumen degradability and intestinal digestibility can alter the absorbed AA profile relative to the diet.

Although this tool is currently limited to research applications because it relies on plasma EAA data, it represents an important step toward making the EAA clustering model more accessible to future studies. If consistent relationships between clusters and performance are observed across studies, future developments could integrate on-farm data sources such as dietary composition to predict cluster membership. In this way, the application could evolve into a practical tool for precision nutrition, supporting the grouping of cows with similar EAA concentration and profile and enabling dietary adjustments that target specific groups to improve animal performance.

AA Profile Cluster Predictor

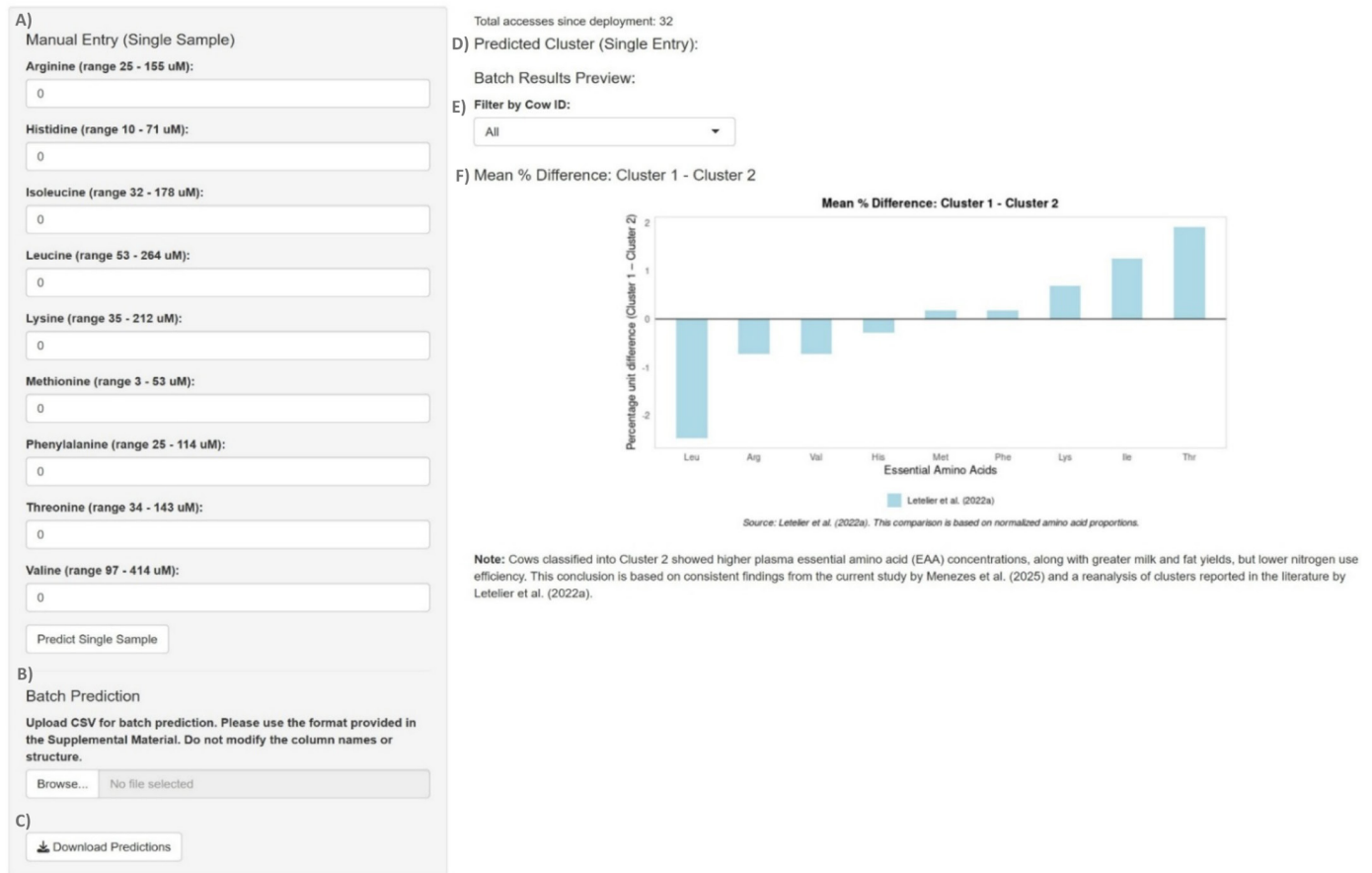


Figure 7. Cluster prediction interface of the web-based classification tool. (A) Manual entry: input plasma concentrations of 9 EAA for one cow and predict its cluster as an illustrative example. (B) Batch prediction: upload a CSV file with multiple cows' EAA concentration for batch classification. (C) Download predictions: download the batch prediction results as a CSV file. (D) Predicted cluster (single entry): displays the cluster result for the single sample entered in section A. (E) Filter by cow id: filter the batch prediction results table by specific cow ID. (F) Mean % difference: cluster 1 – cluster 2: average percentage differences in normalized EAA profile between clusters (from Letelier et al., 2022a and user data).

CONCLUSIONS

This study validates the application of plasma EAA clustering at the individual cow level using a supervised machine learning model, confirming that previously published findings of group-level associations between EAA concentration and profile and dairy cow performance hold true for individual-cow data. Although milk component concentrations remained unaffected by cluster, cows classified into cluster 2 exhibited greater plasma EAA concentrations, DMI, milk yield, fat yield, and lactose yield, as well as reduced NUE compared to cows classified into cluster 1. These results suggest that plasma EAA concentration and profile may influence metabolic pathways relevant to production and N utilization. Our findings support the potential of EAA clustering for precision nutrition research and highlight the need for future studies to evaluate the effects of different dietary inter-

ventions on EAA concentration and profile to establish causality with dairy cow performance and NUE.

NOTES

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opportunity provider and employer. We acknowledge the work of Wendy Radloff (US Dairy Forage Research Center, Madison, WI) for assistance with the laboratory analysis. Data will be made available upon reasonable request. Supplemental material for this article is available at <https://doi.org/10.17605/OSF.IO/Z94YP>. Machine learning methodology was used as part of the statistical analysis of the data. No other forms of artificial intelligence were used at any other stage of this research. No human or animal subjects were used, so this analysis did not require approval by an Institutional Animal Care and Use Committee or Institutional Review Board. All procedures involving animals in the prior study of Letelier et al. (2022b) were approved by the University of Wisconsin–Madison Institutional Animal Care and Use Committee (protocol no. A005043-R01-A03). The authors have not stated any conflicts of interest.

Nonstandard abbreviations used: 3-MH = 3-methylhistidine; BCAA = branched-chain AA; C = cluster; Cov = covariate period; LOOCV = leave-one-out cross-validation; LOSOCV = leave-one-study-out cross-validation; MY = milk yield; NUE = N use efficiency; Par = parity; PUN = plasma urea N; TMG = trimethylglycine; W = weeks (1–4 and 5–8).

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